

Spherical Geometry

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1 S^2 and its Isometries

To start we will talk about what is S^2 and its certain components and rules.

Definition 1.1. S^2 is the 2-dimension unit sphere sitting in $\mathbb{R}^3$ and centered at the origin.

By this definition S^2 is hollow and 2 dimensional as there is no points contained in the sphere just the outside shell is what exists. Along with that seen by the equation of S^2 it must always be centered at the origin, otherwise this is no longer in our definition of S^2 and is just a 2 dimensional sphere in $\mathbb{R}^3$

There are a variety of ways to measure distances in our case. The most classical one for distances in $\mathbb{R}^3$ is given by this first equation:

$$D_{\mathbb{R}^3} = \sqrt{x^2 + y^2 + z^2} \quad (1)$$

This is the more typical way to measure distances in $\mathbb{R}^3$, and will be useful for determining distances between multiple points and generalizing the formula to lower dimensions. However when working with the sphere it is often more practical to talk in terms of distances about the curve of the sphere. We can use the equation:

$$D_{S^2} = r\theta \quad (2)$$

Where theta is the angle of measurement from two points along the circle that they intersect, and r is the radius on the same circle. Another useful trick is to think of the sphere as made up of circles have varying radius. One classification for a type of circle is a great circle where the $r = 1$. One note to consider is that when the two points lie along a great circle the radius is equal which is just the angle between the points.

Definition 1.2. Let $\text{Isom}(S^2)$ be the set of invertible functions from $S^2 \rightarrow S^2$ such that the distances between two points is equal to the distance of the two points after the transformation for all pairs of points.

Any Isometry of S^2 must preserve the distance between any two points before and after the transformation. An example of this would be reflecting about a plane that lies on the Z-Axis. This transformation preserves distance between two points before and after, and maps the points to their corresponding points on either the top hemisphere to the bottom hemisphere.

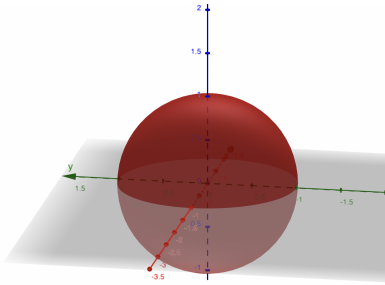


Figure 1: Reflection Isometry

However, a non-example to this would be Figure 2, where the plane does not pass through the origin and while the distances are preserved the reflection created by this plane results in a sphere that is not centered at the origin and by Definition 1.1 it no longer preserves S^2 .

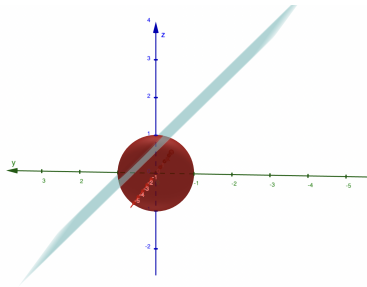


Figure 2: Non-Example

1.1 Classifications of the Isometries of S^2

Theorem 1. *Any Isometry of S^2 is at most the composition of three reflections.*

Proof omitted.

Recalling our definition of an isometry (Definition 1.2) there are three types of transformations that preserve this definition:

1. Reflections (1 Reflection) 1
2. Rotations (2 Reflections) 3
3. Glide Reflections (3 Reflections) 4

When combining 2 reflections about two different planes the resulting transformation is a rotation where the two planes intersection lines create an axis for rotation.

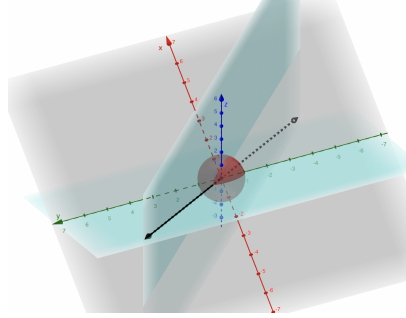


Figure 3: Rotation Isometry

The axis of rotation for the entire sphere lies on the vector $\vec{a}$ which is the line that results from the intersection of the two planes $x + y = 0$, $x + z = 0$

For glide reflections we can use the idea from Figure 3 where we are still rotating about the axis but then we had a reflection about a plane to create the glide reflection, so this isometry is composed of 3 reflections about planes passing through the origin.

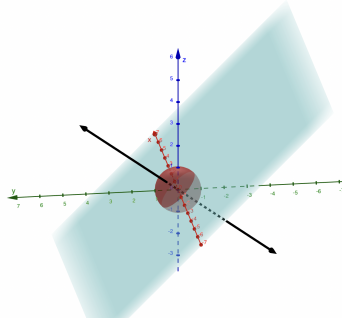


Figure 4: Glide Reflection Isometry

2 Projecting S^2 onto $\mathbb{R}^2$

We will focus on two different projections: Conformal/Stereographic Projections (angle-preserving) and Archimedes' Projections (area-preserving).

(Note: It is not possible to preserve both the angle and area with the same projection (Section 3).)

2.1 Stereographic Projections

Definition 2.1. A Stereographic Projection is a perspective projection of the sphere, through a specific point on the sphere (the pole or center of projection), onto a plane (the projection plane) perpendicular to the diameter through the point. It maps circles on the sphere to circles or lines

on the plane and is conformal, meaning that it preserves angles at which curves meet. However, it does not preserve the area.

Given Definition 1.1, Let $N = (0, 0, 1)$ be the "north pole", and let M be the rest of the sphere. The plane $z = -1$ lies at the bottom of the sphere. For any point on P on M , there is a unique line through N and P , and this line intersects $z = -1$ in exactly one point P' , known as the stereographic projection of P onto the plane.

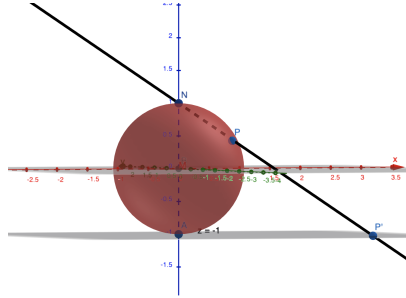


Figure 5: Stereographic Projection

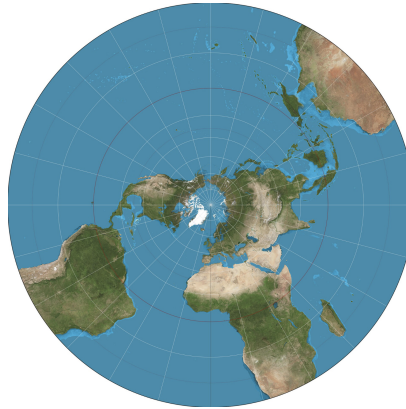


Figure 6: Stereographic Projection of the World

2.2 Archimedes' Projections

Definition 2.2. A method of mapping a sphere onto a cylinder in a way that preserves the area of regions on the sphere. The North/South poles are not included in the projection.

Theorem 2. *Archimedes' area-preserving theorem, also known as Archimedes' Equiareal Map Theorem, states that when a spherical surface is projected onto a circumscribing cylinder, the areas of the regions on the sphere are preserved on the cylinder. However, the angles are not preserved.*

Proof omitted.

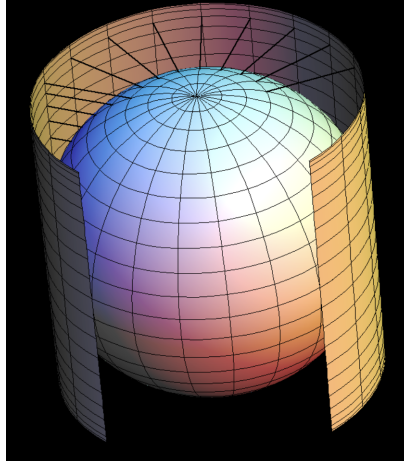


Figure 7: Archimedes' Projection

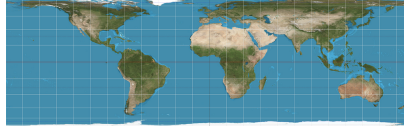


Figure 8: Archimedes' Projection of the World

3 Is there an Isometry $f : S^2 \rightarrow \mathbb{R}^2$?

Theorem 3. *There does not exist an isometry from S^2 to $\mathbb{R}^2$.*

By definition, if one existed, it would necessarily maintain sets of equidistant points. Given a set of equidistant points in S^2 , an isometry should map them to set (of the same size) of equidistant points in $\mathbb{R}^2$. However, it is not implicit (or even true) that size of sets of equidistant points in S^2 and in $\mathbb{R}^2$ are the same size. From prior work done, we understand that the size of these sets that we should investigate is four. Therefore, to disprove the existence of an isometry from S^2 to $\mathbb{R}^2$ it is sufficient to show:

Lemma 1. $\exists p_1, p_2, p_3, p_4 \in S^2$ such that $d_{S^2}(p_i, p_j) = k \forall i, j \in \{1, 2, 3, 4\} : i \neq j, k > 0$

Lemma 2. $\nexists p_1, p_2, p_3, p_4 \in \mathbb{R}^2$ such that $d_{\mathbb{R}^2}(p_i, p_j) = k \forall i, j \in \{1, 2, 3, 4\} : i \neq j$

3.1 Proof of Lemma 1

(Note: The following proof skips some geometric steps that the reader should accept as true). Consider a regular tetrahedron of arbitrary size defined by points A, B, C, and D in $\mathbb{R}^3$. By definition, the

Cartesian distance between each pair of points will be equivalent (A, B, C, and D are equidistant), and each face of the tetrahedron will be an equilateral triangle (created by three of the points). Let us describe face Q_α as the triangle created by all points except α . For example, Q_D is the triangle created by A, B, and C (Figure 9).

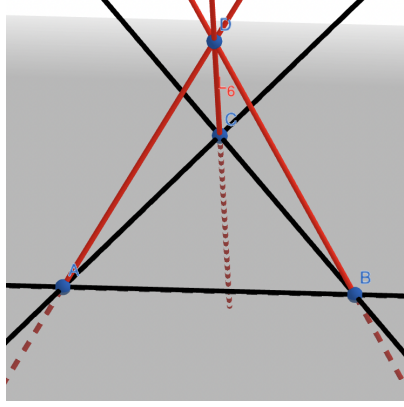


Figure 9: Face Q_D

Define the center of face Q_α (defined by points β, γ, δ) as the component of $\overline{\beta\alpha}$ in the plane containing Q_α . We will call these points C_A, C_B, C_C , and C_D , corresponding to their respective faces. Now, for each point $\alpha \in \{A, B, C, D\}$, create the line $L_\alpha := \overline{\alpha C_\alpha}$. These lines will intersect at a single point O, which we will call the center of the tetrahedron.

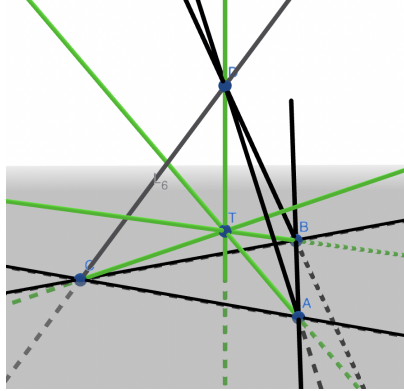


Figure 10: Tetrahedron with L_A, L_B, L_C , and L_D (In Green)

Since each L_α was constructed in the same way and the tetrahedron is symmetric about all planes created by L_α and L_β , it must be the case that T is of equal distance r from A, B, C, and D. Let us scale the tetrahedron so that $r=1$ and move T to the origin of $\mathbb{R}^3$. Note that any rotation about L_α of the tetrahedron maintains the fact that A, B, C, and D are equidistant and $r=1$. Therefore, we

can form the unit sphere (S^2) with rotations about L_A , L_B , L_C , and L_D . Most importantly, since $r = 1$, we know $A, B, C, D \in S^2$.

Claim: Since $A, B, C, D \in S^2$ are equidistant in $\mathbb{R}^3$, they must be equidistant in S^2 .

Proof: Consider two arbitrary points α and β that are the end points of our tetrahedron and in S^2 . Consider the length between α and β on S^2 which we'll call θ . Let us rotate about L_α so that α is fixed and γ moves to β 's old position. We can now easily infer that the length between α and γ on S^2 must also be θ . Use a similar process for the length between α and δ . Since α is an arbitrary choice of A, B, C, or D, we know that each one is the same distance from all other points. Therefore, A, B, C, and D are equidistant in S^2 . $\square$

Therefore, we have found four points in S^2 that are all the same non-zero distance from each other.

3.2 Proof of Lemma 2

Consider two arbitrary points, A and B, in $\mathbb{R}^2$ that are a non-zero distance x apart. If we must find two additional points, C and D, such that all points are equidistant, the following must be true:

- Both C and D must both be a distance of x from A
- Both C and D must both be a distance of x from B
- C and D must be a distance of x from each other

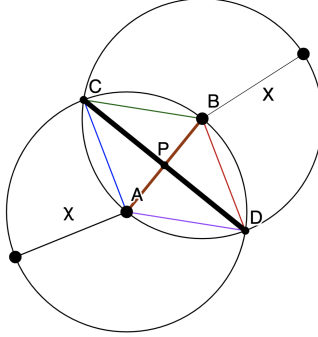


Figure 11: 4 non-equidistant points in $\mathbb{R}^2$

We can visually understand from the first point that C and D must lie on a circle $O_{A,x}$ with radius x centered at A. Similarly, C and D must lie on a circle $O_{B,x}$ with radius x centered at B. Note that since $O_{A,x}$ and $O_{B,x}$ are not identical, they will only have two points of intersection. Therefore, these two points are the only possibilities for C and D. Consider the line from A to B and its midpoint P. The $d(C,D)$ will be equal to the sum of $d(P,C)$ and $d(P,D)$. Since the two intersections of $O_{A,x}$ and $O_{B,x}$ must be symmetric about P, the $d(P,C)$ and $d(P,D)$ must be equivalent, meaning, therefore,

that $d(C,D)$ is simply twice the value of $d(P,C)$. Note that $\overline{PC}$ forms a right triangle with $\overline{PA}$ and $\overline{CA}$ (hypotenuse). Since the $d(P, A) = x/2$ and $d(C, A) = x$, we find the following solution for $d(P,C)$:

$$\begin{aligned} (d(P, C))^2 + (d(P, A))^2 &= (d(C, A))^2 \\ \Rightarrow d(P, C) &= \sqrt{(d(C, A))^2 - (d(P, A))^2} = \sqrt{(x)^2 - (x/2)^2} = \sqrt{3x^2/4} \\ &\Rightarrow d(P, C) = (\sqrt{3}/2)x \end{aligned}$$

So, $d(C,D)$ will be the value of $(\sqrt{3})x$. However, this means that C and D cannot be a distance of x from each other (under the restrictions above).

Therefore, we have proven that there *cannot exist* four equidistant points in $\mathbb{R}^2$.

4 Conclusion

In conclusion, when studying the geometry of the 2-dimensional sphere S^2 sitting in $\mathbb{R}^3$, there are many interesting aspects, including how any isometry of S^2 is the composition of at most three reflections. Theorem 1.

When projecting S^2 onto $\mathbb{R}^2$ there are two main types of projections. Although we cannot preserve both angle and area in one projection, we can preserve them separately with Stereographic Projections 2.1 and Archimedes' Projections 2.2, respectively.

There is no isometry from S^2 to $\mathbb{R}^2$ Theorem 3.